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Paleomagnetism and Tectonic Evolution of the Pan-African Damara Belt, Southern Africa

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Paleomagnetic results are reported from the Nosib, Otavi, and Mulden groups of the Damara Supergroup, a late Precambrian shelf sequence on the southern margin of the Congo craton in Namibia. Three magnetizations were isolated in the Nosib group samples. In order of decreasing blocking temperature they are NQ1 ($n = 6$ sites, $\lambda = 28°N$, $\phi = 323°E$, $\alpha_{95} = 15°$), NQ2 ($n = 7$ sites, $\lambda = 51°S$, $\phi = 213°E$, $\alpha_{95} = 12°$), and NQ3 ($n = 13$ samples, $\lambda = 09°N$, $\phi = 295°E$, $\alpha_{95} = 13°$). Overall precision of all three magnetizations upon tectonic correction suggests that they predate Pan-African (650–450 Ma) folding. Two magnetizations were isolated in the Otavi group samples, above the Nosib in stratigraphic sequence. The DC1 component of possible prefolding age ($n = 4$ sites, $\lambda = 52°S$, $\phi = 186°E$, $\alpha_{95} = 35°$) has been overprinted by the DC2 magnetization ($n = 10$ sites, $\lambda = 55°S$, $\phi = 044°$ E, $\alpha_{95} = 15°$) of probable postfolding age. A single magnetization of probable pre-folding age was isolated in the overlying Mulden Group samples ($n = 6$ sites, $\lambda = 12°S$, $\phi = 090°E$, $\alpha_{95} = 16°$). Together with previously published paleomagnetic data from Africa, the new data showed that no great relative movements have occurred between the Congo and Kalahari cratons during the interval of Pan-African tectonism in the Damara belt (McElhinny and McWilliams, 1977). Continental collision preceded by large relative displacements and closure of a wide ocean (e.g., a Himalayan analog) is effectively ruled out for the Damara belt. We develop an alternative model consistent with the available paleomagnetic and geologic data, which invokes rifting, heating, and stretching of the lithosphere underneath the Damara belt, followed by delamination of the subcrustal lithosphere. Hot asthenospheric material rises to take the place of the detached and sinking lithospheric base, inducing subduction and interstacking of continental crust. The much thickened continental crust is partially melted at depth, intruded by synorogenic and postorogenic granites and finally uplifted and eroded to its present level of exposure. The model is compatible with plate tectonics in that the development of the Damara belt can be broadly compared with modern marginal seas, with the exception that stretching of the lithosphere was not induced by secondary convection above a downgoing slab. Possible causes for stretching are rising mantle plumes or intracontinental distortion within the pre-Damara African plate.

INTRODUCTION

Plate tectonics in the Precambrian is presently a subject of considerable debate, and basically two schools of thought have evolved. One argues that according to uniformitarian principles, global tectonics has evolved along essentially the same lines from early Archean to Phanerozoic times. Since plate interaction is virtually an established fact in modern global tectonics, it must also have been operative in Precambrian times [*Burke and Dewey*, 1972, 1973; *Burke et al.*, 1976, 1977; *Condie*, 1976]. The other school sees a fundamental difference between the tectonic evolution of Precambrian mobile belts and Phanerozoic orogens [*Sutton and Watson*, 1974; *Sutton*, 1977; *Windley*, 1977], concluding that global tectonics may have evolved from an era of predominantly intraplate deformation to the present stage of predominantly plate margin tectonism [*Baer*, 1977; *Kröner*, 1977, 1979*a*].

The African continent has featured in this debate, since it contains a large number of Precambrian mobile belts, some of which have been studied in considerable detail in recent years. Of particular interest is the late Precambrian to early Paleozoic (900–450 Ma) Pan-African system on the eastern and western margins of the continent and in some well preserved interior belts (Figure 1). Some of these mobile belts (e.g., the Mozambique and Zambezi) consist of complexly deformed and metamorphosed pre-Pan-African basement. Here, the question of whether there has ever been an overlying 'geosynclinal' pile which was subsequently eroded during the uplift stage of the Pan-African orogeny or whether the orogeny manifested itself largely through extensive structural reworking and thermal overprinting of older crust [*Clifford*, 1967] is still unresolved. The other type of Pan-African belt (e.g., Damaran, Katanagan, West Congo) evolved from large geosynclinal basins and contains thick late Precambrian to early Paleozoic sediments and only minor volcanics which can be related to typical miogeoclinal and eugeoclinal depositional domains. These latter belts bear close similarity to Phanerozoic orogens, and it has therefore been suggested that they developed through processes of ocean opening and closing [*Burke et al.*, 1976, 1977].

However, field relationships and rock associations do not support this interpretation for all of these belts, and it would appear that the pervasive Pan-African event exhibits features characteristic of both intracratonic and plate margin tectogenesis [*Kröner*, 1977, 1979*a*, *b*, 1980*b*]. Viewed within the global framework of crustal evolution, these settings may indicate that the Pan-African era could reflect a transition from Precambrian ensialic orogeny without large-scale subduction of oceanic crust to Phanerozoic Wilson cycle tectonics [*Kröner*, 1979*a*].

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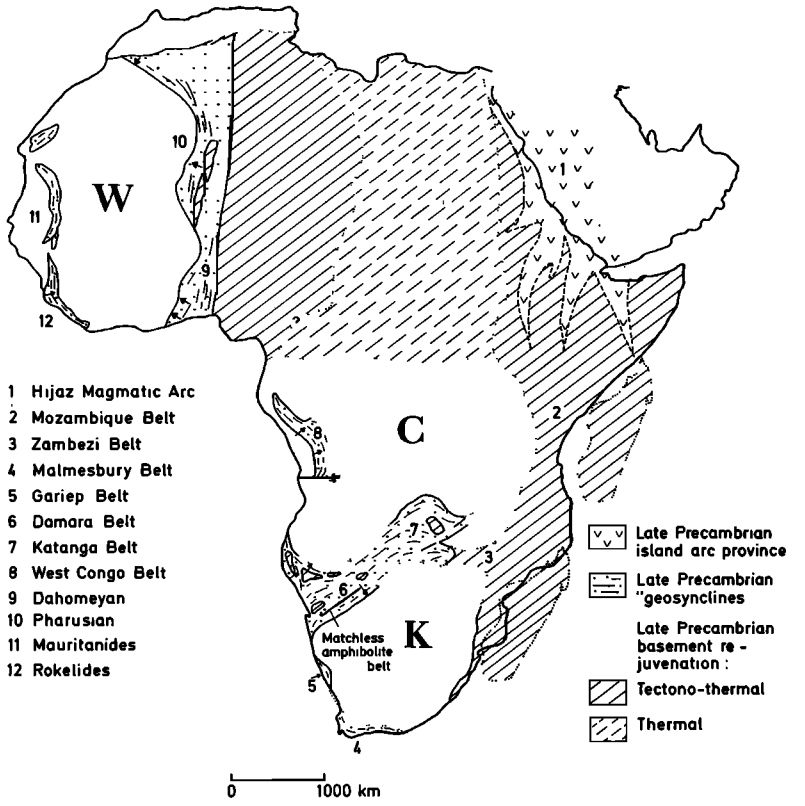


Fig. 1. Generalized structural map of Africa, showing pre-Pan African cratons (W = West African, C = Congo, K = Kalahari), along with areas of Pan African tectonothermal basement reactivation and thermal overprinting. Arrows denote direction of thrusting or nappe transport.

The consequences of this suggestion are considerable, not only for the evolution of the African continent but also for the evolution of the supercontinent Gondwana in Pan-African times, and it is therefore important to investigate each mobile belt individually. An obvious tool is paleomagnetism, since paleomagnetic pole positions from time-equivalent cratonic sequences bordering mobile belts might suggest whether large-scale plate motion preceded collisional orogeny or whether ensialic tectonism had occurred. A considerable amount of new information from a variety of Precambrian sequences in Africa and Australia has led to suggestions that the pre-Phanerozoic parts of these continents had behaved as essentially coherent units during most of Proterozoic time [*Piper et al.*, 1973; *Piper*, 1976; *McElhinny and Embleton*, 1976; *McElhinny and McWilliams*, 1977].

These conclusions were challenged by *Burke et al.* [1976], who argued that it is incorrect to construct Precambrian apparent polar wander paths (APWP) for present-day continents that contain several suture-bounded blocks. Instead, they believe to have identified separate APWP for what they regard as independent plates in southern, eastern, and western Africa, and in their view these paths merge at about 600 Ma,

TABLE 1. Lithostratigraphy of the Damara Supergroup

	Northern Foreland			Center and Southern Foreland		
Age, Ma	Group	Subgroup	Formation	Group	Subgroup	Formation
550–560	Mulden		Owambo	Nama (in parts)		
			Kombat			
			Tschudi*			
Unconformity						
750–840	Otavi	Tsumeb	Hüttenberg	Swakop	Khomas	
			Elandshock			
			Maieberg*			
			Chuos*			Chuos
		Discordance			Discordance	
		Anebab	Auros		Ugab and Kudis	
			Gauss			
			Berg Aukas			
Unconformity						
900–1000		Nosib	Naauwpoort	Nosib		
			Nabis*			

Formation names in the center and south have been omitted. Age data are from *Kröner* [1981*b*].
*Approximate stratigraphic position of samples discussed in this paper.

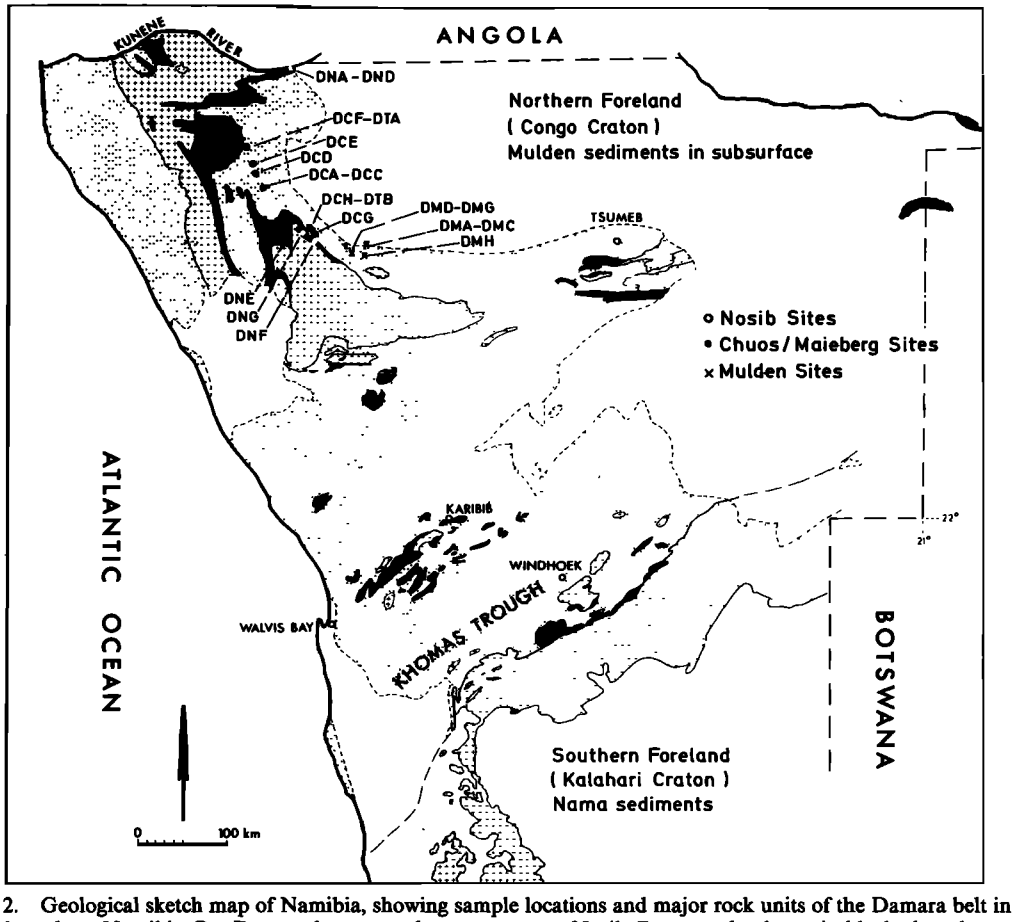


Fig. 2. Geological sketch map of Namibia, showing sample locations and major rock units of the Damara belt in central and northern Namibia. Pre-Damara basement shown as crosses, Nosib Group rocks shown in black, dotted areas underlain by sediments of the Swakop, Otavi, and Mulden Groups (based on map of *Martin and Porada* [1977]).

thereby apparently supporting the collisional character of Pan-African tectonism. *McElhinny and McWilliams* [1977] and *McWilliams* [1981] have discussed this problem at length and rejected the approach of *Burke et al.* [1976] due to the undocumented construction of their APWP for the various cratons.

In this paper we present new paleomagnetic data from northern Namibia and consider the late Precambrian to early Paleozoic APWP for the Congo craton of central Africa and the Kalahari craton of southern Africa (Figure 1), in order to test whether the Damaran-Katangan mobile belt system resulted from collision of previously separate continental plates. The Damara belt is of particular interest because it contains lithologies that can be correlated from one margin to the other [*Martin*, 1965; *Kröner*, 1977; *Martin and Porada*, 1977]. We compare our paleomagnetic results from Damaran strata from the northern shelf domain overlapping the Congo Craton with those of time-equivalent platform sediments and igneous rocks belonging to the Kalahari craton to the south.

Geology and Tectonics of the Damara Belt

As shown in Figure 1, the Damara belt consists of two branches, one trending along the Atlantic coast of Namibia and southern Angola and connecting with the Gariep belt of the Orange River area [*Martin*, 1961; *Kröner*, 1977, 1979*b*], the other crossing the central and northern parts of Namibia in a northeast direction in a broad, nearly 400-km-wide belt between the Kalahari craton to the south and the Congo craton to the north [*Martin*, 1965; *Martin and Porada*, 1977; *Kröner*, 1977]. Relationships between these two branches are still open to considerable debate. It has been suggested that their geometrical configuration resembles that of a triple junction [*Wynne-Edwards*, 1976; *Martin and Porada*, 1977].

The intracontinental branch of the belt has a well developed carbonate shelf facies in the north in contrast to the central and southern regions of the orogen where clastic sediments and only minor carbonates predominate in the lower part, followed by thick flysch-type accumulations [*Martin*, 1965]. The central part of the belt is occupied by medium- to high-grade metamorphic assemblages and voluminous granitoid intrusions. Volcanic rocks are conspicuously rare in the Damara sequence. When present, they consist of sporadically developed but locally thick acid lavas near the base of the sedimentary pile as well as several thin horizons of amphibolite and amphibole schist within the flysch sequence of the southern part of the belt [*Kröner*, 1977]. The stratigraphy, structure, metamorphism, and geochronology are summarized by *Martin* [1965], *Martin and Porada* [1977], *Clifford* [1967], *Hedberg* [1975], *Kröner* [1977, 1980*a*], *Kröner and Clauer* [1979], and *Kröner and Correia* [1981]. Table 1 shows the present lithostratigraphic nomenclature.

The early phase of Damaran evolution was characterized by the formation of several troughs between ~1000 Ma and 800 Ma ago which later widened and subsided to constitute a large basin into which more than 12 km of typical geosynclinal sediments were deposited by about 750 Ma ago [*Martin and Porada*, 1977; *Kröner*, 1980*a*]. In this regard the Damara 'geosyncline' resembles those of Phanerozoic age (although its

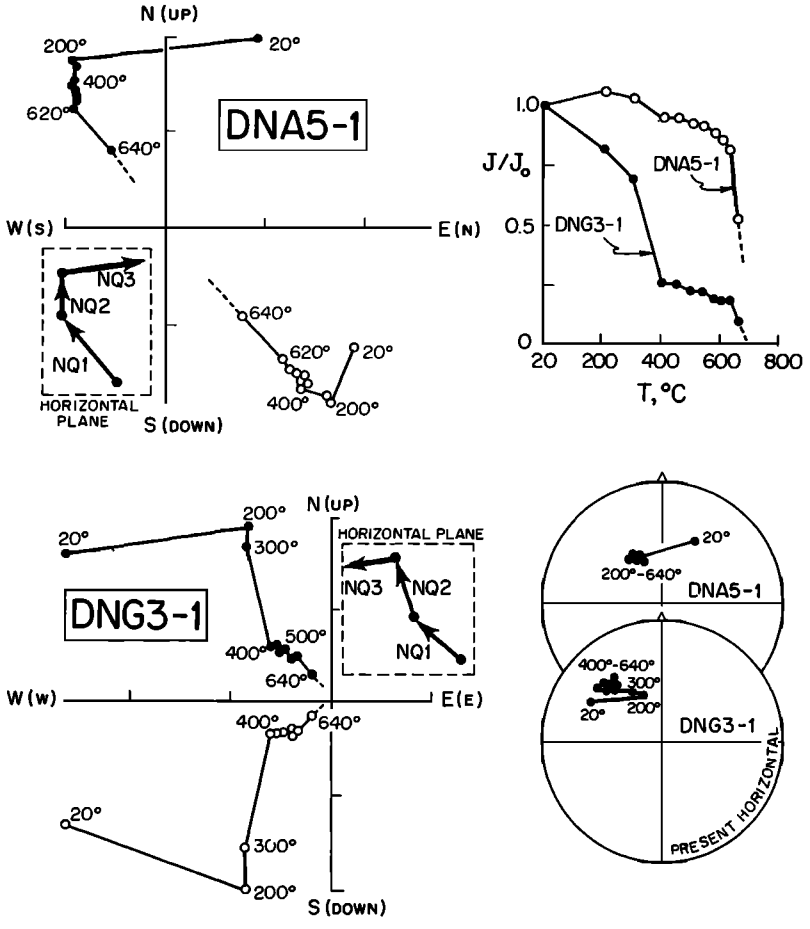


Fig. 3. Examples of thermal demagnetization, Nosib Group samples DNA5-1 and DNG3-1. (Left) Orthogonal vector projection of magnetization vector endpoint on horizontal plane (solid circles) and vertical plane (open circles). Vectors in dashed box illustrate magnetization components in horizontal plane. (Lower right) Equal-angle (stereographic) projection of magnetization directions during thermal demagnetization. Axis ticks: 0.75 × 10^{-3} A/m (DNA5-1), 1.23 × 10^{-3} A/m (DNG 3-1).

development lasted for about 250 Ma), leading *Burke et al.* [1977] to conclude that the Damara belt must have undergone Wilson cycle evolution. There are, however, many features in this well-exposed belt which are incompatible with the Himalaya-type plate collision model, as discussed by *Kröner* [1977] and *Martin and Porada* [1977].

A pronounced asymmetry of the intracontinental branch is evident from facies distribution of sediments, granitoid intrusives, and general structure. To the north, open folding dies out towards the edge of the belt in the Etosha basin and in southern Angola, whereas towards the south, folding and metamorphism become intense and several phases of deformation are recognized. The central part of the belt is deeply eroded, exposing the partly mobilized pre-Damaran basement floor. Age data show that this ~2-Ga-old basement constitutes the southernmost extension of the Congo craton in Namibia [*Jacob et al.*, 1978]. The southern part of the belt is occupied by the Khomas flysch trough (Figure 2) which shows overfolding, shearing, and thrusting of Damaran rocks over the basement in the southern foreland. It contains more than 10 km of metagreywacke with intercalated lenses of amphibolite. The most extensive of these mafic layers is a nearly 300-km-long belt of occasionally pillowed meta-tholeiite with gabbroic or ultramafic intrusives (the Matchless belt) which has been interpreted as obducted ocean floor [*Shackleton*, 1976], although it lacks many characteristics of ophiolites and is clearly interbedded with the greywacke sequence [*Kröner*, 1977].

Martin and Porada [1977] have proposed a model for the Damaran evolution which entails the formation of an aulacogen with subsequent orogenic deformation and metamorphism as a result of crustal detachment and diapiric uprise of an upper mantle asthenolith. *Kröner* [1977, 1979*a*] supported the general concept of an early aulacogen and related the Damaran development to plate tectonic processes involving crustal stretching, incipient continental breakup with formation of a small sialic-floored ocean, and subsequent ocean closure without subduction of oceanic crust. *Wynne-Edwards* [1976] proposed an ensialic spreading model whereby ductile sialic crust underwent extensional and then compressive subhorizontal flow as it moved unilaterally over a spreading center (hot spot) in the upper mantle.

SAMPLING

Samples collected for the present study represent sediments of the Nosib, Otavi, and Mulden groups of the Damara Supergroup. Samples were collected by A.K. and colleagues as part of the South African Geodynamics Project; sampling lo-

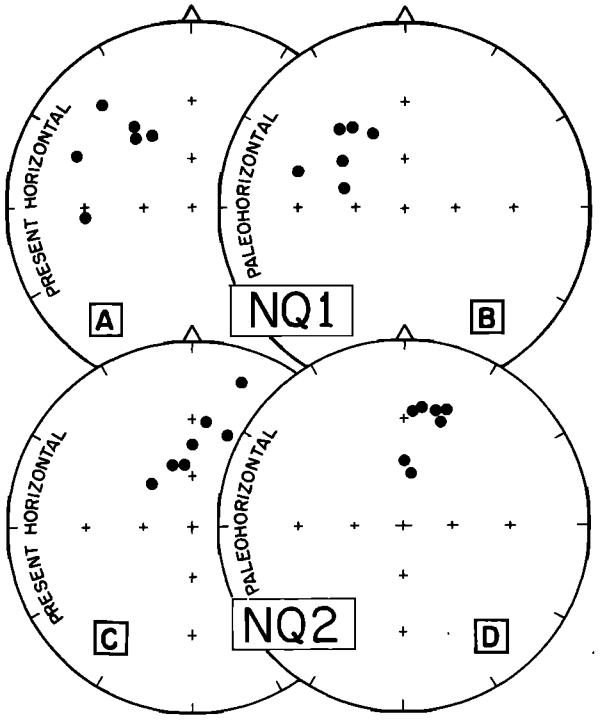


Fig. 4. Site mean magnetization directions, equal-angle projection. Solid symbols: positive inclination. (*a*, *b*) NQ1 component before and after tectonic correction, respectively. (*c*, *d*) NQ2 component before and after tectonic correction, respectively.

calities are shown in Figure 2. The Nosib Group constitutes the basal arenaceous unit of the Damara sequence in Namibia and consists of conglomerate, arkosic quartzite, siltstone, with minor shale and carbonate. Volcanic rocks are found sporadically within the section [*Hedberg*, 1975]. Nosib Group samples were taken at seven sites in feldspathic quartzites of the middle to upper section of the sequence. Although the samples display no outward signs of regional metamorphism, the paleomagnetic results discussed in the following section suggest that some of the rocks may have been partially or totally overprinted during Pan-African tectonothermal activity.

The Nosib sediments are intruded in northern Namibia by the Oas Syenite, dated at 840 ± 13 Ma (Rb-Sr whole rock isochron, $\lambda^{87}Rb = 1.42 \times 10^{-11}$ a^{-1} [*Kröner*, 1980*a*]. In southern Angola, they rest unconformably upon basement granites dated at 1080–1100 Ma [*Kröner and Correia*, 1981]. Regional considerations make it likely that the depositional age of the Nosib Group is between 900 and 1000 Ma. The Chela Group of southern Angola is of similar age and can be correlated with the Nosib Group [*Kröner and Correia*, 1981].

The Otavi Group constitutes a typical carbonate shelf facies and rests paraconformably or unconformably upon the Nosib Group. It also rests unconformably upon the Oas Syenite and is therefore younger than 840 ± 12 Ma. Otavi Group samples for the present study come from the Chuos Formation, one of the most widespread later Precambrian glaciogenic sequences in southern Africa [*Kröner and Rankama*, 1973; *Kröner*, 1975]. The Chuos Formation consists of unbedded ferruginous mixtite, iron-rich quartzite, and minor shale. In the Kaokoveld region of northwestern Namibia, the glaciogenic origin of the Chuos Formation has been documented by *Martin* [1965] and *Hedberg* [1975]. The mixtite is overlain conformably by carbonate rocks of the Maieberg Formation containing stromatolites; samples from basal parts of the carbonates were taken at two localities. The rocks have a visible cleavage, but the metamorphic grade appears to be subgreenschist facies.

An upper age limit for the Chuos mixtite is given by an Rb-Sr model depositional age of 550–560 Ma for the overlying Mulden Group, as discussed below. An additional age constraint can be inferred from the fact that the Chuos mixtites were involved in intense metamorphism in the central Damara belt (Figure 1), dated at 651 ± 34 Ma [*Kröner et al.*, 1978]. Felsic lava flows interfingering with Chuos mixtites yield minimum U-Pb zircon ages of between 742 and 770 Ma (Burger, cited by *Kent* [1973]). Thus an age of about 760 Ma appears most likely for deposition of the Chuos Formation and of glaciation in southwestern Africa.

The Mulden Group represents the youngest unit of the Damara Supergroup and constitutes a typical molasse facies of the Damara belt. It rests uncomformably upon the folded Otavi Group and consists of conglomerate, quartzite, siltstone, and shale. The lower Mulden is a redbed sequence which bears many similarities to the Fish River Formation of the Nama Group, a platform succession on the Kalahari craton to the south (Figure 2).

Samples for the present study were obtained from thickly bedded maroon and grey siltstones which are representative of the lower part of the Mulden Group. Although no metamorphic effects are outwardly visible in the samples, the development of chlorite flakes seen in thin section indicates that some samples may have undergone a very low-grade metamorphism. Rb-Sr studies of Mulden Group shales [*Clauer and Kröner*, 1979] suggest a model depositional age for the Mulden of 550–560 Ma, followed by two distinct low-grade thermal events of regional extent at 537 ± 7 and 457 ± 12 Ma, respectively.

TABLE 2*a*. Nosib Group NQ1 Magnetization

Site	*T*	*n*	*R*	α_{95}	In Situ		Corrected	
					Dec	Inc	Dec	Inc
DNA	500	5	4.868	14	330	43	337	43
DNB	500	4	3.743	28	324	34	327	35
DNC	500	4	3.969	9	320	39	307	44
DND	500	4	3.972	9	293	22	289	27
DNE	350	4	3.961	11	264	30	289	52
DNF	500	4	3.981	7	137	−53	201	−34
DNG	500	4	3.928	14	318	18	320	32

T, temperature, *N*, number of sites; *n*, number of samples; *R*, vector resultant; α_{95}, radius of circle of confidence at $P = 0.05$; Dec, Inc, declination and inclination; *k*, precision parameter; Lat, Long, paleomagnetic pole position; *dp*, *dm*, semiminor and semimajor axes of polar error ellipse at $D = 0.05$.

TABLE 2b. Nosib Group NQ2 Magnetization (Subtracted Vector Directions)

Site	T Range	n	R	α_{95}	In Situ		Corrected	
					Dec	Inc	Dec	Inc
DNA	300–450	5	4.833	16	006	29	009	25
DNB	300–450	4	3.857	21	019	33	021	29
DNC	300–500	4	3.735	29	017	10	016	25
DND	200–500	3	2.966	16	359	41	001	52
DNE	200–350	4	3.971	9	342	50	005	27
DNG	300–450	4	3.985	7	352	51	009	59

See footnotes to Table 2a.

PALEOMAGNETIC RESULTS

Measurements

Samples were collected as field drilled cores using both solar and magnetic compass for orientation. Three or four cylindrical specimens 25 mm in diameter and 22 mm in height were obtained from nearly all the samples. Orientation information in the form of azimuth and dip of cores and block samples is estimated to be correct to within ±2°. Measurement of remanent magnetization was done using one of two magnetometers: a DIGICO slow speed (~7 Hz) flux gate spinner magnetometer, or a 3.8-cm access 2-axis superconducting quantum interference device (SQUID) magnetometer interfaced to the DIGICO processor. A practical lower intensity limit of about 2.0 m 10^{-5} A/m was realized for remanence measurements on the cryogenic instrument. Thermal demagnetization experiments were performed in air in a nonmagnetic furnace capable of obtaining temperatures in excess of 700°C surrounded by a 10-coil feedback controlled field cancellation system. Residual dc magnetic fields during cooling periods were kept to less than 2 nT. Alternating field (af) demagnetization was done in a three-axis tumbler mounted within a large coil capable of obtaining peak fields of over 210 mT; dc field cancellation within the af apparatus was kept to within 10 nT during demagnetization. Transfer of partially demagnetized specimens from the particular demagnetization device to the magnetometer was done in specially constructed μ-metal shields inside which dc magnetic fields were of the order of 50 nT. Specimens were loaded in the magnetometer inside small Helmholtz coil sets surrounding the instrument which cancalled the ambient field within the sample handling region to within 100 nT.

Nosib Group

The NRM (natural remanent magnetization) directions from samples of the Nosib Group were scattered but had a trend toward shallow WNW-ESE declination with shallow inclination. To investigate the possibility of thermal overprinting in these rocks, a detailed thermal demagnetization program with at least 10 heating steps was employed. At least one and often two specimens from each sample were included in the pilot demagnetization experiments. Examples of behavior of the remanent magnetization vector during demagnetization are shown in Figure 3.

In the initial 20°–200°C demagnetization step, a large component with a shallow inclination and easterly or westerly declination was removed from 13 samples at 3 sites. In the 200°–450°C range, a second component with a northerly declination and moderate positive inclination was consistently removed from 24 samples at 6 sites. Magnetization vectors at the latter 6 sites became directionally stable at somewhat higher temperatures (550°–630°C), converging on a northwesterly direction with a moderate positive inclination. This high T_B (blocking temperature) component is called the NQ1 magnetization, while the intermediate and low T_B components are called NQ2 and NQ3, respectively. Upon treatment at temperatures greater than 640°C, magnetization intensity dropped sharply and directional instability ensued.

Resolution of the various components by alternating field (af) demagnetization was not directly possible due to overlapping coercive force (H_c) spectra, although in some cases the stable NQ1 magnetization could be isolated by af treatment alone. At all but one site (DNF), two or more components of magnetization were observed. At site DNF, only a single component was observed which appears to be reversed in polarity with respect to the NQ2 magnetization (Figure 4), and it has been included with NQ2 in Table 2b.

Overall precision of the NQ1 and NQ2 magnetizations increases considerably upon structural correction (Table 2). However, none of the increases are significant at the 95% confidence level. This is a result of (1) the relatively small number of sites and (2) the small difference in local tectonic attitude between sampling sites. There is therefore only a strong suggestion but no unambiguous proof that the relatively well determined NQ1 and NQ2 magnetizations were acquired before Pan-African folding, which began 650 Ma ago.

The more poorly determined NQ3 magnetization direction is not nearly as reliable as NQ1 and NQ2. Considering the

TABLE 2c. Nosib Group, Mean Directions and Pole Positions

Magnetization	N	n	R	k	α_{95}	Dec	Inc	Lat	Long	dp, dm
NQ1*	6	25	5.597	12.4	20	308	33	27°N	317°E	13, 23
[NQ1]†			5.761	20.9	15	312	40	28°N	323°E	11, 18
NQ2	6	24	5.727	18.3	16	004	36	52°S	200°E	11, 19
[NQ2]			5.800	24.9	14	011	36	51°S	211°E	09, 16
NQ2 – DNF	7	28	6.557	13.5	17	359	40	49°S	193°E	12, 20
[NQ2 – DNF]			6.770	26.1	12	012	36	51°S	213°E	08, 14
NQ3		13	11.473	7.9	16	102	−07	10°N	292°E	08, 16
[NQ3]			11.925	11.2	13	102	−13	09°N	295°E	07, 13

See footnotes to Table 2a.
*Site DNF not included in calculation of mean direction.
†Brackets refer to directions after tectonic correction.

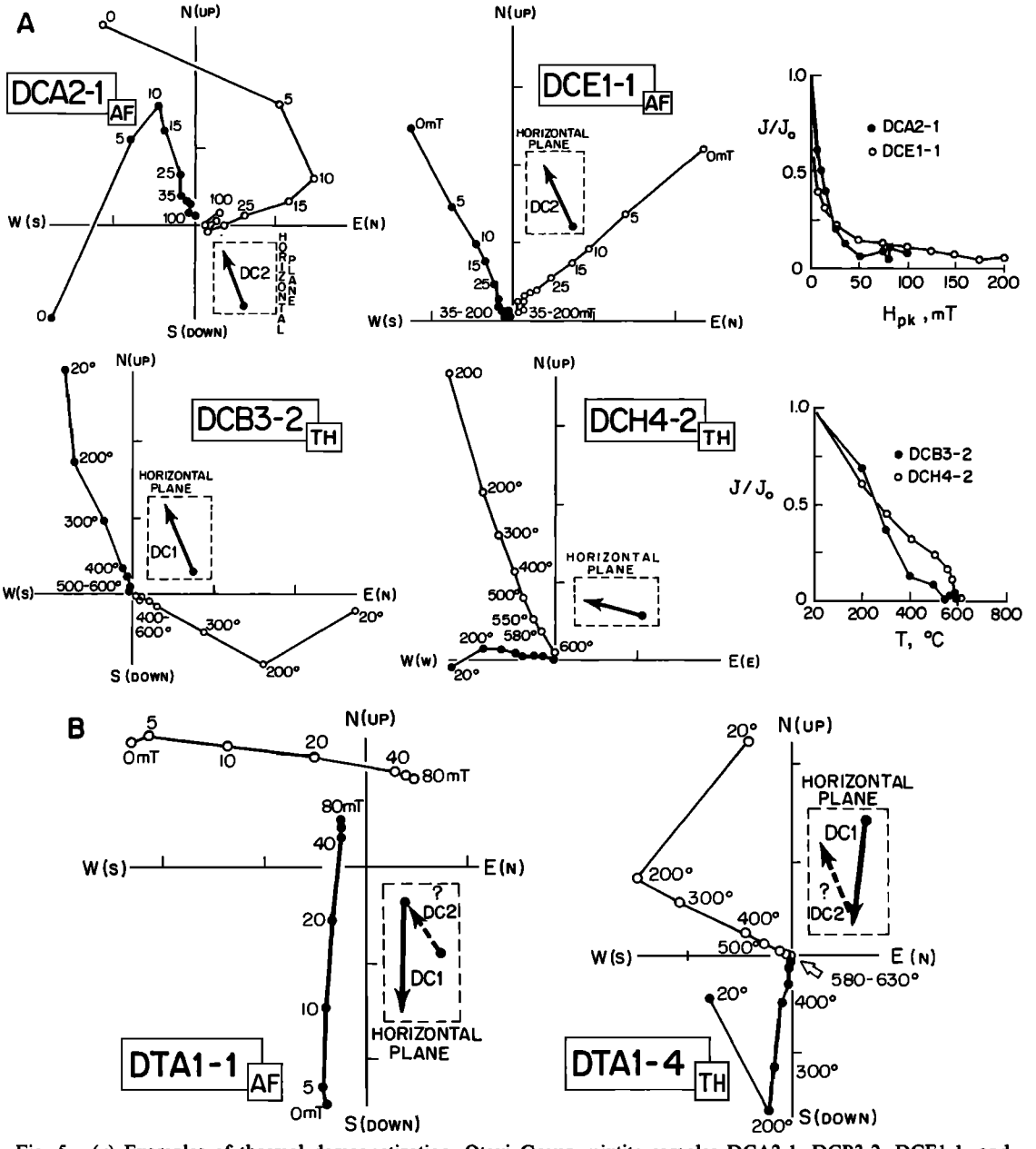


Fig. 5. (*a*) Examples of thermal demagnetization, Otavi Group mixtite samples DCA2-1, DCB3-2, DCE1-1, and DCH4-2: conventions as in Figure 3. Axis ticks: 1.39×10^{-3} A/m (DCA2-1), 8.25×10^{-3} A/m (DCB3-2), 24.7×10^{-3} A/m (DCE1-1), and 64.6×10^{-3} A/m (DCA4-2). (*b*) Examples of alternating-field (DTA1-1) and thermal (DTA1-4) demagnetization of two specimens from the same sample, Otavi Group carbonates. Axis ticks: 3.52×10^{-3} A/m (DTA1-1) and 11.2×10^{-3} A/m (BTA1-4).

narrow temperature range over which it is observed, it is indeed possible that NQ3 is not a 'pure' component but could be contaminated by other low T_B components. Additional samples are required to determine the nature of this magnetization. The fold test information cannot be used to test relative magnetization age sequence. In the absence of post-diagenetic chemical remagnetization the simple interpretation that lower blocking temperatures are exhibited by younger magnetizations would suggest that the magnetization age sequence is NQ1-NQ2-NQ3. As discussed in a following sec-

TABLE 3*a*. Otavi Group DCl Magnetization

Site	T	n	R	α_{95}	In Situ Dec	In Situ Inc	Corrected Dec	Corrected Inc
DCA	420	3	2.927	24	305	33	316	42
DCB	400	4	3.989	6	323	26	333	30
DTA	400	4	3.923	15	188	−14	198	−18
DTB	500	4	3.938	13	141	−64	202	−32

See footnotes to Table 2*a*.

TABLE 3b. Otavi Group DC2 Magnetization

Site	Treatment	n	R	α_{95}	In Situ Dec	In Situ Inc	Corrected Dec	Corrected Inc
DCC	25 mT	2	1.988		337	−36	323	−34
DCD	100 mT	2	1.995		064	−60	061	−76
DCE	125 mT	2	1.999		311	−53	283	−27
DCF	560°C	4	3.757	27	109	59	113	33
DCG	560°C	4	3.359	11	313	−71	261	−36
DCH	560°C	4	3.830	23	193	−83	229	−26

See footnotes to Table 2a.

tion, this simple model (which is probably quite realistic in the case of purely thermal effects in a low-grade metamorphic terrane) fits the available data rather well. To summarize, our interpretation of the results from the Nosib sediments is that (1) NQ1 is the oldest magnetization (primary?), (2) NQ1 was subsequently thermally overprinted by NQ2 and possibly by NQ3, and (3) it is likely that NQ1 and NQ2 predate Pan-African folding and tectonism in the Damaran belt.

Otavi Group

NRM directions from Otavi group sites were widely scattered and random at the 95% confidence level; af and thermal demagnetization experiments conducted on at least one specimen per sample revealed a complex magnetization system for many of the mixtite and carbonate samples. The majority of mixtite samples from sites DCC–DCH exhibited only a single consistent magnetization (DC2) with a characteristic northwesterly (southeasterly) declination and negative (positive) inclination, whether during af (DCE1-1, Figure 5a) or thermal (DCH4-2, Figure 5a) demagnetization. The DC2 magnetization at these sites has T_B distributed between 200° and 600°C and H_c generally greater than 150–200 mT.

At mixtite sites DCA and DCB, this same DC2 component was observed as a subtracted vector direction during af demagnetization between 10 and 35 mT (DCA2-1, Figure 5a). During thermal demagnetization of samples from these sites, however, the DC2 component was not observed, and the characteristic magnetization has a northwesterly declination with a positive inclination (DCB3-2, Figure 5a). This thermally isolated component, labeled DC1, has T_B distributed in the 200°–600°C range in the mixtite samples.

Demagnetization characteristics of the Otavi Group carbonate samples were different from those of the mixtite samples, although the same DC1 and DC2 magnetizations were observed. During af demagnetization (DTA1-1, Figure 5b), the DC1 magnetization exhibits H_c in the 5–80 mT range, while the DC2 magnetization has much higher H_c (af demagnetization in fields greater than 80–100 mT did not change the magnetization vector). During thermal demagnetization, the order of magnetization removal is reversed, with a magnetization parallel to DC2 removed in the low-temperature steps (DTA1-4, Figure 5b). The example of Figure 5b shows this low T_B magnetization removed in the initial 20°–200°C step; blocking temperatures of the DC2 magnetization in the carbonate samples were as high as 400° in some samples.

The DC1 magnetization was observed at 4 sites, and its overall precision is increased only slightly upon tectonic correction (Table 3d and Figures 6a and 6b). The DC2 magnetization was observed at 6 sites where it was the only coherent vector direction, as well as a low T_B magnetization at the 4 sites where DC1 was observed. Upon tectonic correction, the overall precision of DC2 decreases significantly (at 93% level), indicating a postfolding age for DC2 (Figures 6c and 6d). In view of their relative blocking temperatures and the fold test information, a magnetization age sequence DC1–DC2 is preferred. In this interpretation, the DC1 magnetization predates the latest phase of Pan-African folding but is not necessarily primary. It is quite possible that DC1 could date from an early Pan-African, pre-Damaran thermal episode, possibly the intense event at 651 ± 34 Ma [*Kröner et al.*, 1978] in the central Damara belt. The DC2 magnetization is interpreted as a younger magnetization which has partially (sites DCA, DCB, DTA, DTB) or totally (sites DCC–DCH) overprinted the preexisting DC1 magnetization.

Mulden Group

NRM directions from the Mulden Group samples were scattered, but a trend towards shallow easterly (westerly) directions is suggested. During pilot thermal demagnetization studies of one or more specimens from each sample, magnetization directions at 6 sites converged upon a common shallow easterly (westerly) direction after being heated to temperatures of 520°–620°C, confirming the trend observed in the NRM directions. During the course of the thermal demagnetization experiments, no systematic subtracted vector directions were observed (Figure 7), and thus only a single magnetization (DM) was isolated from the Mulden Group sediments. The DM magnetization is characterized by high coercive forces with MDF (median destructive field) in many cases exceeding 180 mT.

After thermal demagnetization a significant improvement in precision compared to NRM is seen in the overall mean DM direction when referred to present horizontal (Figure 8

TABLE 3c. Otavi Group DC3 Magnetization (Subtracted Vector Directions)

Site	Range	n	R	α_{95}	In Situ Dec	In Situ Inc	Corrected Dec	Corrected Inc
DCA	10–35 mT	2	1.947		359	−63	325	−63
DCB	10–35 mT	2	1.988		349	−35	336	−36
DTA	200°–400°C	2	1.970		336	−57	322	−36
DTB	200°–400°C	2	1.999		328	−83	243	−35

See footnotes to Table 2a.

TABLE 3*d*. Otavi Group, Mean Directions and Pole Positions

Magnetization	*N*	*n*	*R*	*k*	α_{95}	Dec	Inc	Lat	Long	*dp, dm*
DC1	4	15	3.555	6.7	38	331	37	42°S	157°E	26, 45
[DC1]			3.620	7.9	35	354	34	52°S	186°E	23, 40
DC2	6	18	5.412	8.5	24	327	−69	48°S	044°E	35, 41
[DC2]			4.957	4.8	34	280	−46	17°S	080°E	28, 43
DC3	4	8	3.811	15.8	24	346	−60	64°S	042°E	27, 36
[DC3]			3.488	5.9	42	307	−49	41°S	080°E	37, 55
DC2 + DC3	10	26	9.180	11.0	15	336	−65	55°S	044°E	20, 24
[DC2 + DC3]			8.338	5.4	23	291	−48	27°S	080°E	20, 30

See footnotes to Tables 2*a* and 2*c*.

and Tables 4*a* and 4*b*). Precision of the DM magnetization improves markedly upon structural correction, but due to shallow dips and similar bedding attitude at sites in the sampling area the improvement is not significant at the 95% confidence level.

Magnetization vectors at sites DMA and DMH reached quite stable endpoints during pilot demagnetization, but these stable directions were quite different from the DM direction and from each other (Tables 4*a* and 4*b*). Site DMA yielded a mean direction very similar to the N3 overprint from sediments of the Nama Group to the south of the Damara belt [*Kröner et al.*, 1980]; in fact, a VGP (virtual geomagnetic pole) falls within the N3 polar confidence ellipse at 95% (Figure 9). A widespread Cambrian thermal event had been postulated as the cause of the N3 overprinting, and the isotopic studies of *Clauer and Kröner* [1979] would suggest that the N3 and DMA magnetizations may date from such an event at 537 ± 7 Ma. The well-grouped DMH site mean direction is unlike any magnetization observed in Damara Supergroup or Nama Group strata and is also unlike any younger Phanerozoic magnetization direction for Southern Africa.

APW Paths and Relative Motion of the Congo and Kalahari Cratons

The available paleomagnetic data for southern Africa representing the interval from about 1300 Ma to about 450 Ma ago, together with the new data from this study, are summarized in Table 5 and plotted with different symbols for different structural blocks of Africa in Figure 9. Since the focus of this paper is an estimation of the relative motion between the Congo and Kalhari cratons and the evolution of the Damara belt, we have included data from those cratons only in Table 5 and Figures 9*a* and 9*b*. The problem of paleomagnetism and Pan-African orogenesis in the general context of Gondwana is discussed in a separate paper [*McWilliams et al.*, 1981]. The general APWP trend from ~1300 Ma to ~650 Ma is similar to that suggested by *McElhinny and McWilliams* [1977], *Briden et al.* [1980], and *McWilliams* [1981]. The latest Precambrian to early Paleozoic part of the path for southern Africa of Figure 9 differs from the path suggested by *Kröner et al.* [1980] in that for times before about 530 Ma ago, the path has been reversed in polarity, resulting in a slightly longer total path length. Both suggested paths are viable alternatives; ambiguity arises from the lack of reliable data in the interval separating poles DM, N2, and NQ3 from N3, SJ, H1, KL, and DG. This results in an approximately 90° gap in the APWP in early Cambrian times which, when combined with the uncertainty in polarity of older poles, permits two possible interpretations. Fortunately, this uncertainty does not affect the interpretation of the data in terms of the tectonic evolution of the Damara belt, as poles from both the Congo and Kalahari change hemispheres when polarity is reversed.

It is apparent from Figure 9 that a single APWP can be constructed which contains all of the available data from the Congo and Kalahari cratons in the interval~1000–450 Ma. This path can be constructed without violating any of the applicable magnetization age constraints, and poles of approximately equivalent magnetization age from both cratons are in good agreement at 4 points along the proposed APWP: ~950

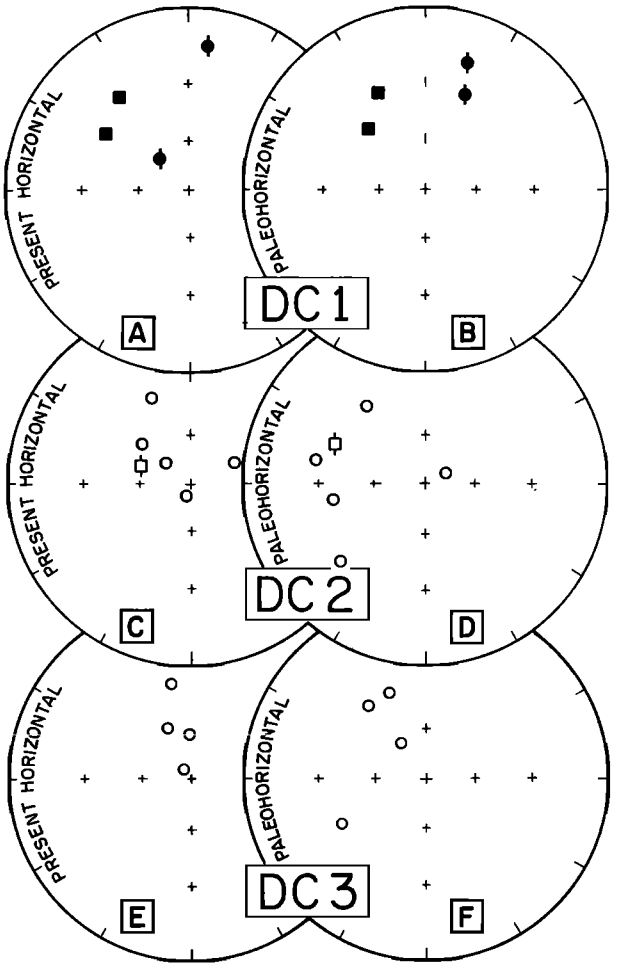


Fig. 6. Site mean magnetization directions: conventions as in Figure 4. (*a*, *b*) DC1 component before and after tectonic correction, respectively. (*c*, *d*) DC2 component (observed vector) before and after tectonic correction, respectively. (*e*, *f*) DC3 component (subtracted vector directions from samples which exhibit DC1 component) before and after tectonic correction, respectively. Symbols with superimposed vertical bars represent directions which have been reflected through the origin to plot on upper hemisphere.

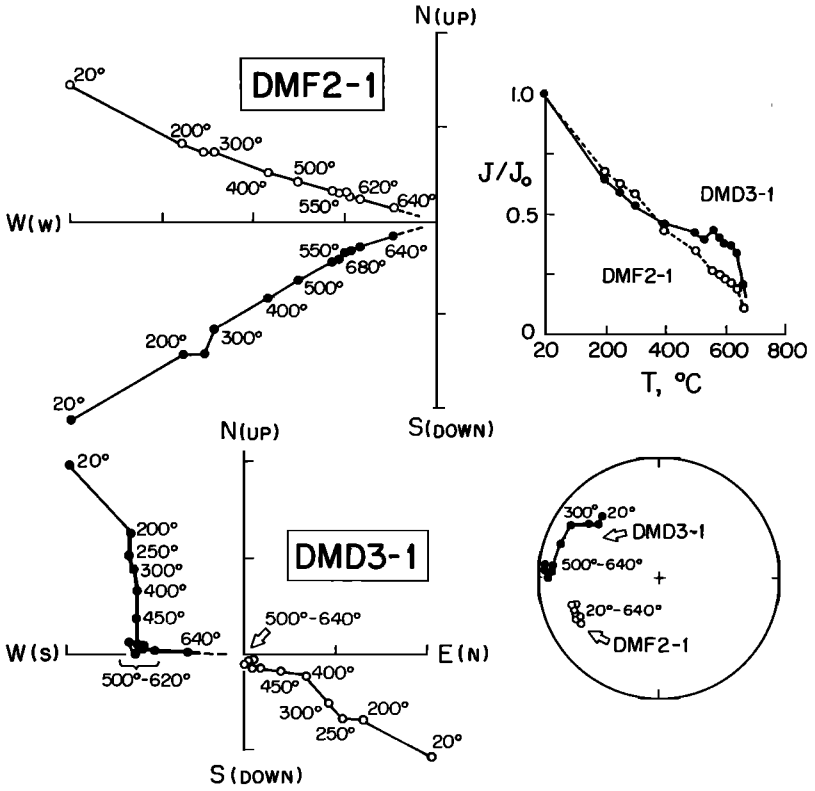


Fig. 7. Examples of thermal demagnetization, Mulden Group samples DMF2-1 and DMD3-1. Conventions as in Figure 3. Axis ticks: 0.65×10^{-3} A/m (DMF2-1) and 0.26×10^{-3} A/m (DMD3-1).

Ma, ~850 Ma, ~600 Ma, and ~570 Ma. These points represent times before and during episodes of Pan-African orogenesis in the Damara belt, and thus it is clear that the paleomagnetic data are in accord with any model for Damaran evolution which does not postulate large-scale relative motion between the Congo and Kalahari cratons.

'Large-scale' refers here to relative motion in excess of the uncertainty limits of the paleomagnetic data. For simple latitudinal changes, these uncertainties average approximately 10° or about 1100 km. It is important to remember that the absence of differential APW between cratons does not immediately preclude differential cratonic motion. For example, when the paleomagnetic pole and the Euler pole describing differential cratonic motion are in close proximity, considerable motion can occur which will escape detection by the paleomagnetic method. Conversely, the presence of differential APW does not directly imply that large-scale differential cratonic motion has occurred, since simple cratonic rotations about local Euler poles can produce significant differential APW with only small displacements when the craton occupies low paleolatitudes. However, when long and well-defined APWP segments from two cratons can be brought into coincidence, a unique determination of relative position can be safely assumed. Further problems and limitations in detecting relative motion between crustal blocks using Precambrian paleomagnetic data are discussed by *McWilliams* [1981].

With this in mind, a reexamination of Figure 9*b* illustrates that poles DM and N2 (from the Congo and Kalahari cratons, respectively) are in an unfavorable position to detect an important component of possible relative motion between the two cratons. Keeping the Congo craton fixed, the Damara belt strikes approximately WSW-ENE; obviously, the Kalahari craton cannot have come from the north, so that on a collisional model, we would expect that a significant component of the motion of the Kalahari craton would be northwards. But poles DM and N2 are approximately 90° to the east in low latitudes, situated such that this northward component might be undetectable. Although this might at first appear to be a difficult problem, most workers would agree that in any collisional model, much of the postulated Pan-African convergence had already occurred by 570 Ma, since the main structural event in the Damara belt probably took place about 650 Ma ago [*Kröner*, 1980*a*, *b*]. An analysis of older data is required to test whether the two cratons were totally independent crustal fragments which later collided.

Poles in the ~750- to 600-Ma portion of the path (e.g., N1,

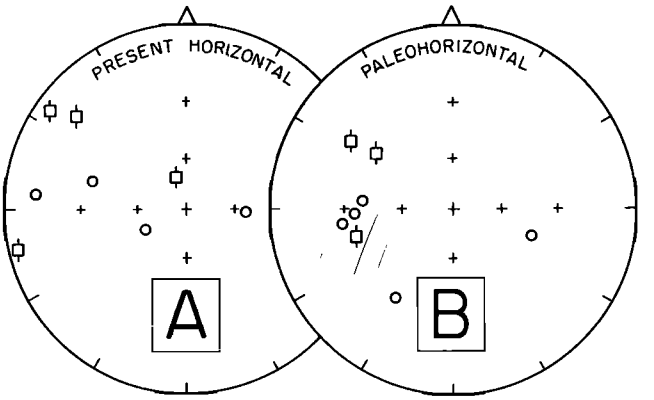


Fig. 8. Site mean magnetization directions, Mulden Group: conventions as in Figure 4. Squares with superimposed lines have been reflected through the origin to lie on the upper hemisphere.

TABLE 4a. Mulden Group Site Mean Directions

Site	T	n	R	α_{95}	In Situ Dec	In Situ Inc	Corrected Dec	Corrected Inc
DMA	580	4	3.971	9	093	−55	107	−41
DMB	580	4	3.845	21	286	−34	275	−38
DMC	580	4	3.989	6	275	−11	262	−28
DMD	580	4	3.604	36	125	05	124	23
DME	620	4	3.963	10	270	−11	267	−34
DMF	620	4	3.893	18	076	03	074	33
DMG	620	4	3.838	22	129	14	126	36
DMH	620	4	3.984	7	241	−61	213	−31

See footnotes to Table 2a.

NQ2, MZ) are in excellent position to detect motion normal to the regional strike of the Damara belt. Again, keeping the Congo craton fixed, they lie directly to the south, and thus northward relative motion of the Kalahari craton would produce differential APW on approximately a 1:1 scale. No such differential motion is apparent from the data. Here again, proponents of collisional models might argue that a considerable amount of convergence may have already occurred by 650–600 Ma in a collisional orogenic model, and thus we step back further in time to check for pre-650-Ma relative cratonic motion. Such relative motion is indicated by existing geological data and is incorporated in our tectonic model presented below.

On any Wilson-cycle model for the Damara belt (with the exception of a special case discussed below), the Congo and Kalahari cratons must have been independent crustal fragments at least between about 800 and 700 Ma ago, occupying different positions on the globe and (in all probability) in relative motion. They should therefore have different pre-700-Ma APW paths. Inspection of Figure 9 shows that this is not the case; indeed, wherever poles of approximately equal age are available from both cratons, they are in good agreement. Poles on the APWP at about 1000 Ma are in an excellent position to detect relative motion, but no such motion is seen. This is incompatible with any postulated large-scale relative motion and subsequent collision.

The suggested APWP (Figures 9a, 9b) illustrates that the available data from both cratons are not distributed evenly in time. Characteristically, the path is defined in some intervals by data from one craton, then by overlapping data from both cratons, and then by data from the other craton. We must therefore consider the question, What of those intervals along the path fow which the path is defined by data from a single craton? Only one special set of circumstances could make the observed paleomagnetic data compatible with any postulated large-scale relative motion and collision: if the relative motion between cratons occurred in an opening-closing, cyclical fashion such that the cratons were apart only when data are available from one craton and always came back together in their previous configuration for the intervals for which overlapping data exist. Although such a scenario is a possible interpretation of Wilson-cycle tectonics in southern Africa, we argue that to postulate such a restricted special condition is ad hoc by comparison to the much simpler interpretation indicated directly by the data.

A Model for the Evolution of the Damara Belt

There is no longer much doubt that the formation of orogenic belts, even in Precambrian times, is closely linked to the motion of lithospheric plates or segments of lithospheric plates. Phanerozoic orogens can often be successfully traced to Wilson-cycle processes, and there are basically 3 settings that permit the formation of linear belts: (1) ocean opening and closing followed by continental collision, (2) continuous subduction of oceanic crust under a continental margin (Andean-type evolution), and (3) back-arc basin opening during ocean-crust subduction and eventual closure following continental collision [*Dennis*, 1976; *Dennis and Jacoby*, 1980]. The present debate centers on the question whether these or similar settings were also generated during Precambrian times or whether Archean and Proterozoic plate motions produced different results.

The Damara belt is particularly interesting in this regard, since it lacks many of the features which would be expected if any of the settings above were applicable for its evolution. For instance, it contains no recognizable ophiolites, blueschist assemblages, or volcanic rocks of calc-alkaline affinity, and many of the granitoid rocks have been produced by anatexis of the geosynclinal fill or parts of the pre-Damara basement. These features rule out an Andean-type evolution, and the limitations imposed by the paleomagnetic data reported above restrict movement of the Congo and Kalahari cratons with respect to each other. A back-arc basin development would be in line with this limitation, since it would probably go undetected with presently available paleomagnetic information. However, a variety of geological arguments are in conflict with this model:

1. The Damara belt is now~400 km wide, and the original depositional basin must have exceeded ~500 km in width, a figure far too large if compared to most modern marginal basins, except, perhaps, for the Sea of Japan.

TABLE 4b. Mulden Group, Mean Directions and Pole Positions

Magnetization	N	n	R	k	α_{95}	Dec	Inc	Lat	Long	dp, dm
DM*	6	24	5.600	12.5	20	283	−14	15°N	283°E	10, 20
[DM]*			5.715	17.6	16	278	−34	13°N	270°E	10, 18

See footnotes to Tables 2a and 2c.
*Sites DMA and DMH omitted from calculation of mean direction.

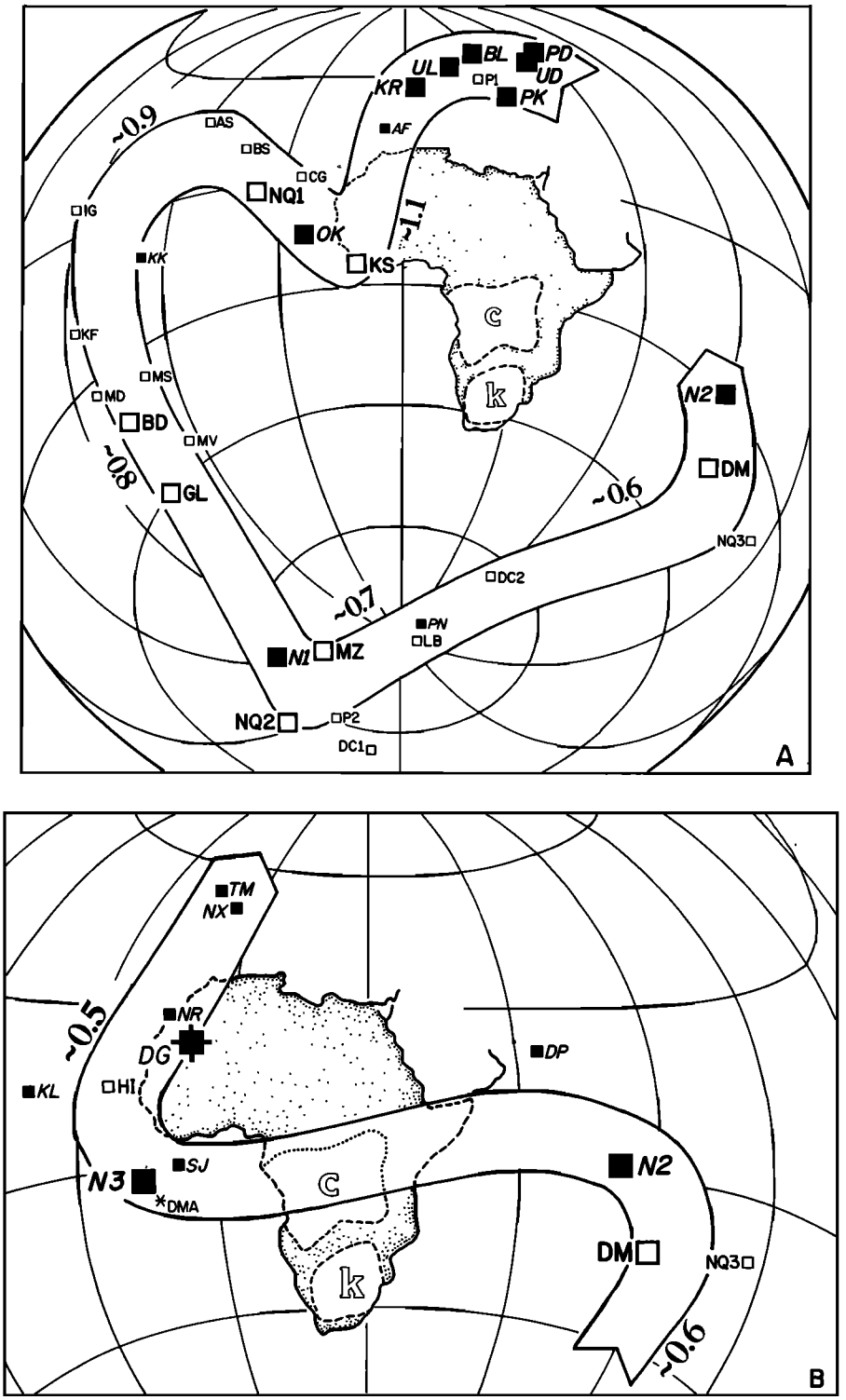


Fig. 9. (*a*) 1.2 to 0.6-Ga APWP for Congo (c) and Kalahari (k) cratons. Open (solid) squares represent poles from Congo (Kalahari) craton. Large symbols: $R \geq 5$ (Table 5). Equal-area projection; Africa in present coordinates; grid lines at 30° intervals. Symbols as in Table 5. (*b*) 0.6-Ga Ordovician APWP for Congo (c) and Kalahari (k) cratons. Conventions as in Figure 9*a*. Approximate ages in Ga.

2. The back-arc basin model implies a volcanic arc on one side of the basin which is not found in the Damara belt.

3. There is no forearc basin, nor is there any indication of subduction which would be required to generate a marginal basin.

We therefore conclude that this model is not applicable to the Damara belt, although it is likely that the mechanism of marginal basin formation may be similar to that inferred for the Damara geosyncline, as argued below.

What remains as a possible model is a continental collision following limited ocean opening and little or no subduction of oceanic crust during ocean closure (Figure 10). Such settings

TABLE 5. Late Precambrian–Early Paleozoic Paleomagnetic Data for Southern Africa

Symbol	Formation	Possible Magnetization Age*	R†	Reference‡
		Congo Craton		
KS	Kisii Series, Kenya	1200–1010 (K)	5	GJ14/522
CG	Chela Group, Angola	1100–1050 (S)	4	*Kröner* [1975]
NQ1	Nosib Group, Namibia	<1010 + 930 (U), >853–827 (R)	6	this paper
BS	Bukoba sandstone, Tanzania	<1200, >1000 (S)	4	GJ14/543
P1	Plateau Series, Zambia	<AS (S)	2	GJ14/560
IG	Ikorongo Group, Tanzania		3	GJ14/537
KF	Kigonero flags, Tanzania	>GL (S)	2	GJ14/544
MS	Malagaresi sandstone, Tanzania	1200–800 (S)	2	GJ14/542
MD	Mbala dolerite, Zambia	940–680 (K)	3	GJ14/558
BD	Bukoba dolerite, Tanzania	836–776 (K)	6	GJ14/541
NQ2	Nosib Group, Namibia	<NQ1, 679–660 (R)	6	this paper
DC1	Otavi Group, Namibia	684–618 (R)	4	this paper
MV	Manyovu redbeds, Tanzania	<GL (S)	4	GJ14/545
GL	Gagwe lavas, Tanzania	843–783 (K)	6	GJ14/540
MZ	Mbozi complex, Tanzania	773–713 (K)	6	GJ14/539
LB	Lake Buanji Series, Tanzania	<1350 (S)	4	GJ14/538
DC2, 3	Otavi Group, Namibia	<DC1 (M)	4	this paper
NQ3	Nosib Group, Namibia	<NQ2 (M)	3	this paper
DM	Mulden Group, Namibia	560–550 (R)	6	this paper
DMA	Mulden Group, Namibia (VGP)	<DM, 544–530 (R)	2	this paper
HI	Hook intrusives, Zambia	506–473 (R)	4	GJ09/132
P2	Plateau Series, Zambia	Lower Paleozoic (?)	2	GJ14/561
		Kalahari Craton		
PK	Premier Kimberlite, S. Africa	<1250, >1130–1100 (R)	5	GJ10/197 and *Ito et al.* [1978]
PD	Post-Waterberg dolerite, Botswana	~1250 (R)	5	GJ08/155
UD	Umkondo dolerite, Rhodesia	~1140 (R)	5	GJ08/154
BL	Barby lavas, Namibia	1280–1250 (R)	5	GJ14/533
UL	Umkondo lavas, Rhodesia	~1115 (R)	5	GJ08/154
KR	Koras Group, South Africa	1196–1160 (R)	5	*Briden et al.* [1980]
AF	Auborous Formation, Namibia	<1250 (R, S)	4	GJ14/535
OK	O'Okiep intrusives, S. Africa	1090–1050 (U)	6	GJ14/532
KK	Klein Karas dykes, Namibia	~880 (K)	3	GJ14/536
NL	Lower Nama Group, Namibia	<718–654, >566–540 (R)	5	*Kröner et al.* [1980]
PN	Pre-Nama dykes, S. Africa	723–583 (K)	4	GJ14/530
KL	Klipheuval Formation, S. Africa	Pc/∈ (S)	3	*Creer* [1973]
NZ	Upper Nama Group, Namibia	∈1 (S)	6	*Kröner et al.* [1980]
DP	Doornpoort Formation, Namibia	PA2 (M)	4	GJ14/529
SG	Sijarira Group, Rhodesia	Pc/∈ (S)	3	GJ12/149
N3	Nama overprint, Namibia	PA2 (M), 540–520 (K)	5	*Kröner et al.*[1980]
TM	Table Mt. sandstone, S. Africa	O (S)	2	GJ04/32
NX	Blaubeker overprint, Namibia	500–440 (R, U, K)	3	*Kröner et al.* [1980]
		Mozambique Belt		
NR	Ntonya Ring Structure, Malawi	641–593 (R), 520–480 (K)	5	GJ09/137
		Damara Belt		
DG	Damaran granites	458–500 (R)	6	*Corner and Henthorn* [1978]

*K, K/Ar; R, Rb/Sr; U, U/Pb; S, stratigraphic estimate based on field relationships and known isotopic age information; M, estimated from paleomagnetic information, e.g., fold tests, relative blocking temperatures.

†Reliability sum (P + A of *McElhinny and Embleton* [1976]).

‡GJ prefix refers to *Geophysical Journal* paleomagnetic data compilations.

have been proposed by *Kröner* [1977, 1979*a*] and *Martin and Porada* [1977] but were regarded with scepticism [e.g., *Windley,* 1981] or rejected outright, since they imply that most of the crustal shortening in the deformed geosyncline was induced by underthrusting and/or interstacking of continental crust, or, specifically, continental subduction.

Continental subduction, for a long time anathema to theoretical plate tectonicians, has now been demonstrated in the Himalayas [*Toksöz and Bird,* 1977; *Bird,* 1978] and in the Alps [*Hsü,* 1979] and has been postulated in the western United States under the Colorado Plateau [*Bird,* 1979]. *Molnar and Gray* [1979] demonstrate that significant fractions of the lower continental crust up to several hundred kilometers in length may be subducted if they can become detached from their upper part. Also, the gravitational force acting on the descending old and dense subcrustal lithosphere may pull parts of the lower continental crust with it until this pull is counterbalanced by the high buoyancy of the subducted crustal slab [*Hargraves,* 1981]. Continental delamination or lithospheric split has probably occurred beneath the northern Appennines in the Tertiary and may be a common feature of the continental collision mechanism [*Reutter et al.,* 1980; *Bally,* 1981]. All that is

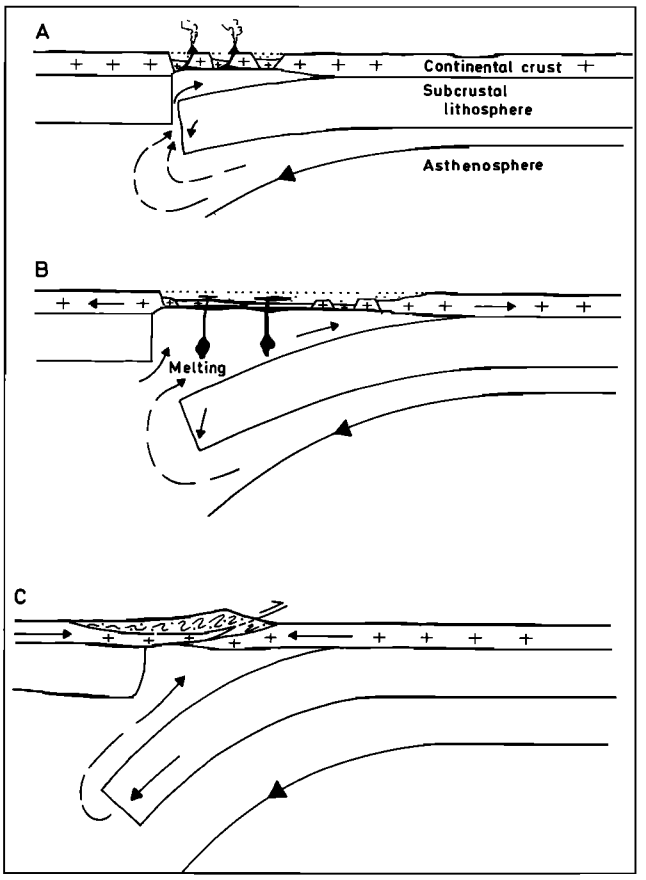


Fig. 10. Cartoon depicting three stages in the evolution of an ensialic orogenic belt through (*a*) initial rifting and delamination of subcrustal lithosphere, (*b*) crustal stretching and formation of a geosyncline basin floored by thin sialic crust, and (*c*) orogenic deformation through crustal interstacking or A-subduction causing horizontal shortening and intense deformation of the sedimentary fill. Note that no subduction of ocean crust is involved.

needed are fractures in the lithosphere to free the subducting segment [*Bird*, 1979]. Figure 10 diagrammatically demonstrates the rifting, delamination, and crustal interstacking mechanism which leads to 'ensialic' orogenic deformation without subduction of oceanic crust.

The similarity of many features in the Damara belt with those observed in the Alps suggest a broadly similar evolution (J. Dewey and A. Kröner, manuscript in preparation, 1981) and on the basis of the paleomagnetic constraints presented here, new geochronological data [*Kröner*, 1980*a*], and previously published geodynamic interpretations [*Kröner*, 1977; *Martin and Porada*, 1977] we propose the following model as depicted in Figures 10 and 11:

1. Fracturing in the pre-Damara basement of the combined Congo and Kalahari cratons began about 950–1000 Ma ago and induced the development of several graben systems into which the clastic sediments of the basal Nosib Group were deposited.

2. Further rifting and crustal stretching produced localized acid volcanic rocks and alkaline intrusives about 840–900 Ma ago and led to widening of the early graben-structures to broad basins (Figure 10*a*).

3. Continued heating of the lower crust by the evolving asthenospheric flow under the rifted lithosphere induced ductile spreading and considerable thinning of the continental crust. The embryonic basins merged and developed into a wide ensialic geosyncline by about 800 Ma ago.

4. We now speculate that further stretching caused fracturing of the entire subcrustal lithosphere and beginning of spontaneous lithospheric detachment and delamination with attendant lateral spreading of rising asthenosphere at or near the base of the crust-mantle boundary (Figure 11*b*). The southern part of the Damara basin now deepened and was rapidly filled with thick greywacke accumulations of the Khomas Subgroup. At the same time, mantle-derived tholeiitic lavas with chemical affinities to ocean-floor basalts [*Kröner*, 1979*b*] were extruded into the basin, perhaps signifying almost complete crustal failure at that stage. The fact that these lavas are interbedded with the Khomas greywackes and contain no evidence of lateral spreading (no sheeted dykes) suggest to us that they do not represent oceanic crust and entire crustal separation has therefore probably not occurred.

A provisional age of 775 ± 33 Ma for the lavas [*Kröner*, 1980*a*] dates that stage of the geosynclinal evolution where the basin reached its greatest width, and it is important to note that our paleomagnetic data are in accord with the above model in indicating a common APWP for the Congo and Kalahari cratons at that time.

5. Delamination and sinking of the dense subcrustal lithosphere induces subduction and interstacking of continental crust. Since no oceanic crust was generated, none could be subducted, and therefore no calcalkaline magmatism developed. Instead, the geosynclinal fill was subjected to compressive deformation, finally leading to high-level thrusting and nappe tectonics along the southern margin and fo' 'and of the belt (Figure 11*c*).

Subcrustal heating following delamination and lateral spreading of asthenospheric material, combined with crustal thickening as a result of continental subduction, caused partial melting in the crust and at the crust/mantle boundary, producing the family of synorgenic to postorogenic granites of the Damara belt between ~650 and ~500 Ma ago.

6. The effects of continental subduction are similar to those observed after continental collision in modern orogenic belts, and, above all, they cause considerable uplift and erosion to compensate for crustal thickening. In the Damara belt this phase was reached about 650–570 Ma ago when molasse sequences were deposited in the north (Mulden Group) and in the south (Nama Group).

We believe that the above model is consistent with the plate tectonic concept and the limitations imposed on the relative motions of the Kalahari and Congo cratons during Pan-African times. We feel that the Damara basin was not an aulacogen as suggested by *Martin and Porada* [1977]. Its development may be comparable to modern marginal seas with the exception that crustal stretching was not induced by secondary convection above a subduction zone but perhaps by a mantle plume or by intracontinental distortion within the large pre-Damara African plate, similar to the movements which produced the Adelaide geosyncline and the Amadeus basin of Australia [*McWilliams*, 1981].

Paleomagnetic data continue to support the suggestion that continental subduction and ensialic orogeny as proposed above have been widespread in the Proterozoic, and we wish to emphasize that this model is not anti-plate-tectonic as some have argued but merely represents a variation of the global

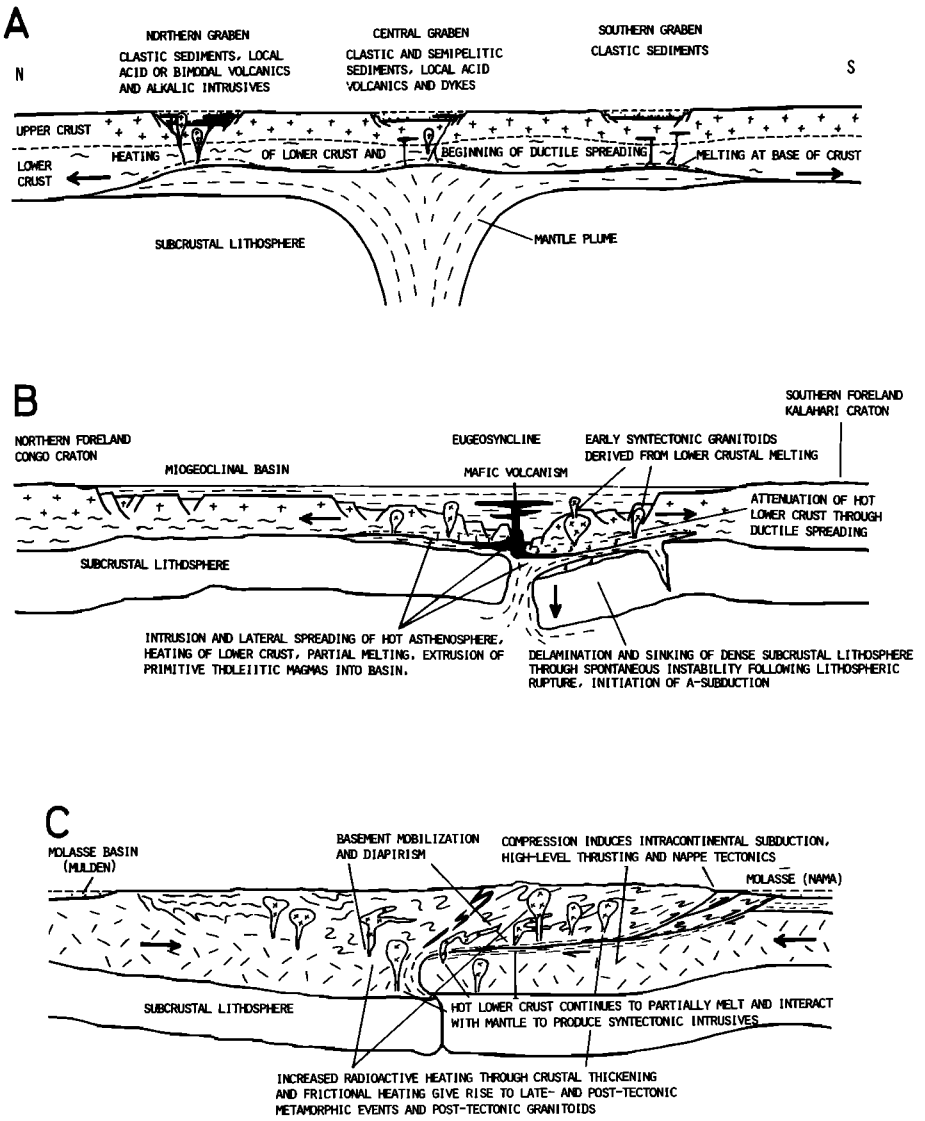


Fig. 11. Simplified and schematic sections showing suggested evolution of the apparently ensialic Pan African Damara belt through crustal thinning, delamination, and intracrustal A-subduction. For detailed explanation of model, see text and *Kröner* [1981*a*, *b*].

scheme of plate interaction which may still operate today as shown by the evolution of the Alps [*Hsü*, 1979].

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